



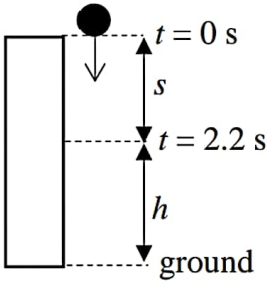
SKEMA JAWAPAN DAN PEMARKAHAN PRA PR
FIZIK SEMESTER 1, SESI 2024/2025

NO.	ANSWER SCHEME																	
1.(a)(i)	$m = 8 \text{ mg} = 8 \times 10^{-3} \times 10^{-3} = 8 \times 10^{-6} \text{ kg}$																	
1.(a)(ii)	$V = 50 \text{ mm}^3 = 50 \times (10^{-3})^3 = 50 \times 10^{-9} \text{ m}^3$																	
1.(b)	<table><tr><td></td><td>x-component (km)</td><td>y-component (km)</td></tr><tr><td>S₁</td><td>2</td><td>0</td></tr><tr><td>S₂</td><td>$3.5\cos40^\circ$ =2.68</td><td>$-3.5\sin40^\circ$ =-2.25</td></tr><tr><td>S₃</td><td>$2.5\cos64^\circ$ =1.10</td><td>$2.5\sin64^\circ$ =2.25</td></tr><tr><td>$\sum S$</td><td>$\sum S_x = 5.78$</td><td>$\sum S_y = 0$</td></tr></table> $S = \sqrt{S_x^2 + S_y^2}$ $S = \sqrt{5.78^2 + 0^2}$ $S = 5.78 \text{ km}$ Direction : To the East (to the right)				x-component (km)	y-component (km)	S₁	2	0	S₂	$3.5\cos40^\circ$ =2.68	$-3.5\sin40^\circ$ =-2.25	S₃	$2.5\cos64^\circ$ =1.10	$2.5\sin64^\circ$ =2.25	$\sum S$	$\sum S_x = 5.78$	$\sum S_y = 0$
	x-component (km)	y-component (km)																
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NO.	ANSWER SCHEME
2.(a)(i)	$t_1 = (0 - 6) \text{ s}$ $v_1 = \text{gradient of } s\text{-}t \text{ graph}$ $v_1 = \frac{-4 - 20}{6 - 0} = -4 \text{ m s}^{-1}$ 2 / 11 $t_2 = (6 - 12) \text{ s}$ $v_2 = \frac{4 - (-4)}{12 - 6} = 1.33 \text{ m s}^{-1}$ <u>Graph v versus t</u> Label v, t and SI unit : 1 mark Shape : 1 mark
2.(a)(ii)	$v_{av} = \frac{\sum d}{\sum t}$ $v_{av} = \frac{24 + 8}{12}$ $v_{av} = 2.67 \text{ m s}^{-1}$ OR $v_{av} = \frac{\text{area under } v\text{-}t \text{ graph}}{\text{total time}}$ $v_{av} = \frac{7.94 + 24}{12}$ $v_{av} = 2.67 \text{ m s}^{-1}$
2.(a)(iii)	$v_{av} = \frac{\Delta s}{\Delta t}$ $v_{av} = \frac{4 - 20}{12 - 0}$ $v_{av} = -1.33 \text{ m s}^{-1}$ OR $v_{av} = \frac{\Delta s}{\Delta t}$ $v_{av} = \frac{7.94 - 24}{12 - 0}$ $v_{av} = -1.33 \text{ m s}^{-1}$
2.(b)(i)	$u = 0 \text{ m s}^{-1}$ (dropped), $t = 1.5 \text{ s}$ $v = u - gt$ $v = 0 - 9.81(1.5)$ $v = -14.72 \text{ m s}^{-1}$ OR 14.72 m s^{-1} (downward)

NO.	ANSWER SCHEME
2.(b)(ii)	$u = 0 \text{ m s}^{-1}$ (dropped), $t = 2.2 \text{ s}$ $s = ut - \frac{1}{2}gt^2$ $s = (0)(2.2) - \frac{1}{2}(9.81)(2.2)^2$ $s = -23.74 \text{ m}$ $ s = -23.74 = 23.74 \text{ m}$ Height from the ground, $h = 50 - 23.74 = 26.26 \text{ m}$ 
	TOTAL



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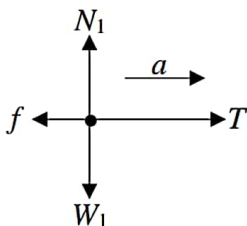
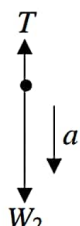
NO.	ANSWER SCHEME	1
3.(a)	$m = 0.015 \text{ kg}, t = 0.012 \text{ s}$ <u>If $u = -20 \text{ m s}^{-1}, v = 10 \text{ m s}^{-1}$</u> $Ft = m(v - u)$ $F(0.012) = 0.015(10 - (-20))$ $F = 37.5 \text{ N}$ OR <u>If $u = 20 \text{ m s}^{-1}, v = -10 \text{ m s}^{-1}$</u> $Ft = m(v - u)$ $F(0.012) = 0.015(-10 - 20)$ $F = -37.5 \text{ N}$ $\therefore$ Magnitude of force, $F = 37.5 \text{ N}$	
3.(b)	$m_1 = ?$ $m_C = ?$ $v = 49 \text{ km h}^{-1}$ $u_T = 50 \text{ km h}^{-1}$ $u_C = 0 \text{ m s}^{-1}$ $\sum P_i = \sum P_f$ 4 / 11 $m_T u_T + m_C u_C = (m_T + m_C)v$ $m_T(50) + (100)(0) = (m_T + 100)(49)$ $m_T = 4900 \text{ kg}$	⋮
3.(c)	$m_1 = 2 \text{ kg}$ $u_1 = 3.5 \text{ m s}^{-1}$ $u_{1x} = 3.5 \text{ m s}^{-1}$ $u_{1y} = 0 \text{ m s}^{-1}$ $v_1 = 2 \text{ m s}^{-1}$ $v_{1x} = -2\cos 50^\circ = -1.29 \text{ m s}^{-1}$ $v_{1y} = -2\sin 50^\circ = -1.53 \text{ m s}^{-1}$ $m_2 = 3 \text{ kg}$ $u_2 = 5 \text{ m s}^{-1}$ $u_{2x} = -5 \text{ m s}^{-1}$ $u_{2y} = 0 \text{ m s}^{-1}$ <u>x-component</u> $\sum P_{ix} = \sum P_{fx}$ $m_1 u_{1x} + m_2 u_{2x} = m_1 v_{1x} + m_2 v_{2x}$ $(2)(3.5) + (3)(-5) = (2)(-1.29) + 3v_{2x}$ $v_{2x} = -1.81 \text{ m s}^{-1}$	

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NO.	ANSWER SCHEME	MARKS
3.(c)	<u>y-component</u> $\sum P_{iy} = \sum P_{fy}$ $m_1 u_{1y} + m_2 u_{2y} = m_1 v_{1y} + m_2 v_{2y}$ $(2)(0) + (3)(0) = (2)(-1.53) + 3v_{2y}$ $v_{2y} = 1.02 \text{ m s}^{-1}$ $v_2 = \sqrt{v_x^2 + v_y^2}$ $v_2 = \sqrt{(-1.81)^2 + 1.02^2}$ $v_2 = 2.08 \text{ m s}^{-1}$	
	TOTAL	

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NO.	ANSWER SCHEME	
4.(a)	$W = 150 \text{ N}, F = 90 \text{ N}$ Constant speed, $a = 0 \text{ m s}^{-2}$ $\sum F_y = 0$ $N - W = 0$ $N = 150 \text{ N}$ $f_k = \mu_k N$ $\sum F_x = 0$ $F - f_k = 0$ $90 - \mu(150) = 0$ $\mu_k = 0.6$	]
4.(b)(i)	<u>Object 1</u>  <u>Object 2</u>  6 / 11 6 corrects : 3 marks 5 corrects : 2 marks 4 corrects : 1 mark (0 – 3) correct/s : 0 mark	

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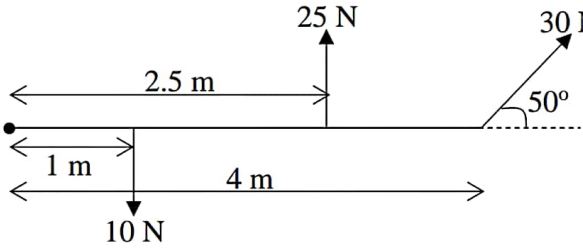
NO.	ANSWER SCHEME	M
.(b)(ii)	$m_1 = 6 \text{ kg}, m_2 = 8 \text{ kg}, \mu_k = 0.22$ $\sum F_y = 0$ $N_1 - W_1 = 0$ $N_1 = m_1 g = (6)(9.81) = 58.86 \text{ N}$ $f = \mu_k N = (0.22)(58.86) = 12.95 \text{ N}$ $\sum F_x = m_1 a$ $T - f = m_1 a$ $T - 12.95 = 6a$ $\sum F_y = m_2 a$ $W_2 - T = m_2 a$ $(8)(9.81) - T = 8a$ $78.48 - T = 8a$ $T = 41.03 \text{ N}$ $a = 4.68 \text{ m s}^{-2}$	
	TOTAL	



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NO.	ANSWER SCHEME	
5.(a)(i)	$r = \frac{d}{2} = \frac{3}{2} = 1.5 \text{ m}, v = 2 \text{ m s}^{-1}$ $a_c = \frac{v^2}{r}$ $a_c = \frac{(2)^2}{1.5}$ $a_c = 2.67 \text{ m s}^{-2}$	1
5.(a)(ii)	$\omega = \frac{2\pi}{T}$ $a_c = r\omega^2$ $a_c = (1.5) \left(\frac{2\pi}{T} \right)^2$ $T = 4.71 \text{ s}$	
5.(b)(i)	$m = 0.15 \text{ kg}, l = r = 0.9 \text{ m}$ $\text{Given } \omega = \frac{1 \text{ rev}}{s} = \frac{2\pi \text{ rad}}{s} = 2\pi \text{ rad s}^{-1}$ $F_c = \sum F_x = ma_c$ $T = mr\omega^2$ $T = (0.15)(0.9)(2\pi)^2$ $T = 5.33 \text{ N}$	8 / 11
5.(b)(ii)	$\text{Given } v = 11.31 \text{ m s}^{-1}$ $F_c = \sum F_x = ma_c$ $T = \frac{mv^2}{r}$ $T = \frac{(0.15)(11.31)^2}{0.9}$ $T = 21.32 \text{ N}$	
	TOTAL	

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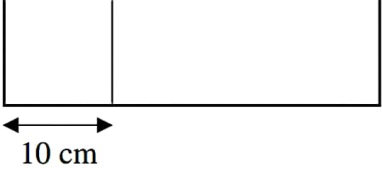
NO.	ANSWER SCHEME
6.(a)(i)	$u = 8.40 \text{ m s}^{-1}, s = 115 \text{ m}, r = \frac{d}{2} = \frac{0.68}{2} = 0.34 \text{ m}$ $u = r\omega_o$ $8.4 = 0.34\omega_o$ $\omega_o = 24.71 \text{ rad s}^{-1}$
6.(a)(ii)	$v = 0 \text{ m s}^{-1}$ $s = r\theta$ $115 = (0.34)\theta$ $\theta = 338.24 \text{ rad}$ $\theta = 338.24 \text{ rad} \times \frac{1 \text{ rev}}{2\pi \text{ rad}}$ $\theta = 53.8 \text{ rev}$
6.(a)(iii)	$\omega^2 = \omega_o^2 + 2\alpha\theta$ $0^2 = 24.71^2 + 2\alpha(338.24)$ $\alpha = -0.90 \text{ rad s}^{-2}$
6.(b)(i)	 $\tau = rF\sin\theta$ $\sum \tau_{cw} = -(10)(1)\sin 90^\circ = -10 \text{ N m}$ $\sum \tau_{acw} = (25)(2.5)\sin 90^\circ + (30)(4)\sin 50^\circ$ $\sum \tau_{acw} = 154.43 \text{ N m}$
6.(b)(ii)	$\sum \tau = \tau_{acw} + (-\tau_{cw})$ $\sum \tau = 154.43 - 10$ $\sum \tau = 144.43 \text{ N m (ecf } \sum \tau_{cw} \text{ and } \sum \tau_{acw})$

TC

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NO.	ANSWER SCHEME	N.
7.(a)(i)	$r = 0.015 \text{ m}, L = 0.3 \text{ m}, T_1 = 60^\circ\text{C}, T_2 = 20^\circ\text{C}$ $k_{\text{copper}} = 400 \text{ W m}^{-1} \text{ K}^{-1}$ $A = \pi r^2 = \pi \times (0.015)^2 = 7.07 \times 10^{-4} \text{ m}^2$ $\frac{Q}{t} = -kA \frac{\Delta T}{\Delta L}$ $\frac{Q}{t} = -400 \times 7.07 \times 10^{-4} \times \left(\frac{20 - 60}{0.3} \right)$ $\frac{Q}{t} = 37.71 \text{ J s}^{-1}$	
7.(a)(ii)	60°C T 20°C  10 cm $\frac{Q}{t} = -kA \frac{\Delta T}{\Delta L}$ $37.71 = -400 \times 7.07 \times 10^{-4} \times \left(\frac{T - 60}{0.1} \right)$ $T = 46.67^\circ\text{C}$ (ecf for $\frac{Q}{t}$)	
7.(b)(i)	$\langle v^2 \rangle = \frac{450^2 + 720^2 + 560^2 + 910^2 + 840^2}{5}$ $\langle v^2 \rangle = 5.14 \times 10^5 \text{ m}^2 \text{ s}^{-2}$	
7.(b)(ii)	$v_{\text{vrms}} = \sqrt{\langle v^2 \rangle}$ $v_{\text{vrms}} = \sqrt{5.14 \times 10^5} = 716.69 \text{ ms}^{-1}$ (ecf for $\langle v^2 \rangle$)	
7.(c)(i)	$W = -100 \text{ kJ}, \Delta U = +55 \text{ kJ}$ $Q = \Delta U + W$ $Q = 55 \text{ k} + (-100 \text{ k})$ $Q = -45 \text{ kJ}$	

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	ANSWER SCHEME
7.(c)(ii)	$\Delta U = -8136 \text{ J}, Q = -9636 \text{ J}$ $Q = \Delta U + W$ $-9636 = -8136 + W$ $W = -1500 \text{ J}$ Work is done on the system.
7.(d)(i)	Process 1 to 2 : Isothermal expansion process Process 2 to 3 : Isochoric/isovolumetric process Process 3 to 1 : Adiabatic compression process
7.(d)(ii)	$Q = -40 \text{ J}$ Isochoric process $\rightarrow W = 0 \text{ J}$, $Q = \Delta U + W$ $-40 = \Delta U + 0$ $\Delta U = -40 \text{ J}$
	TOTAL

